



CRYSTAL PM HOW-TO GUIDE

Connect Crystal Meet rooms to Zoom

Let each room join Zoom meetings through Zoom's Meeting SDK, including meetings hosted by customers and vendors, with no one signing in on the device.

This guide has two halves. The Zoom half is done once for all rooms by the person who administers your Zoom account: two apps in the App Marketplace and one Zoom user per room, about 20 minutes. The device half takes about five minutes per room. Rooms still join only when someone presses Join on the room page.

STEP 1

The SDK app

A Meeting SDK app and its client id and secret.

STEP 2

The token app

A Server-to-Server OAuth app with one scope.

STEP 3

A user per room

Rooms join outside meetings as their own Zoom user.

STEP 4

Put it on the device

The credentials file, the installer, the check.

STEP 5

Join a test meeting

One inside your account, one outside.

BEFORE YOU BEGIN

Access. The Zoom account owner, or an admin with the Develop permission (visible as **Develop** at the top of the App Marketplace) and rights to add users.

Why three parts. Since March 2026 Zoom lets an SDK app join meetings hosted by other Zoom accounts only as a signed-in Zoom user. The SDK app (step 1) joins your own meetings; the token app (step 2) and a Zoom user per room (step 3) let the room join everyone else's.

Secrets. You will collect a client id and secret (step 1) and an account id, client id and secret (step 2). Keep them out of email and chat; they go straight into a file on each device.

1 Create the Meeting SDK app

- Open the App Marketplace at marketplace.zoom.us signed in as the admin, then **Develop › Build App**, choose **General App** and create it. Name it **Crystal Meet rooms**.
- Turn on the Meeting SDK. In the app's **Features** (or **Embed**) section, enable **Meeting SDK**.
- Copy the credentials. On **Basic Information › App Credentials**, copy the **Client ID** and **Client Secret**. The development credentials are fine: the app is used only inside your account and is never published.

You should now see the app under **Manage › Built Apps** with Meeting SDK enabled, and have its client id and secret.

2 Create the Server-to-Server app for tokens

- Build it. **Develop › Build App**, choose **Server-to-Server OAuth**, name it **Crystal Meet room tokens**.
- Copy the credentials from **App Credentials: Account ID**, **Client ID** and **Client Secret**.
- Add the one scope. Under **Scopes › Add Scopes**, search for **token** and add **user:read:token:admin** (View users' tokens). Nothing else is needed.
- Activate the app on the **Activation** page. Tokens only work while it is active.

You should now see the app listed as active, with the account id, client id and secret saved with the step 1 values.

3 Create a Zoom user for each room

- Add the user. In admin.zoom.us go to **User Management › Users › Add Users**. Use a mailbox you can read, for example room1@crystalpm.com, and the **Basic** user type, which is free on a paid account.
- Activate it. Open the invitation email in that mailbox and finish the sign-up.
- Name it after the room. In the user's profile set the display name to the room's name, for example **Room 1**. That is the name hosts see when the room joins.
- Repeat for Room 2 and Room 3.

You should now see one active Basic user per room, each named after its room.

GOOD TO KNOW

The room user never hosts anything and needs no licence. If you skip steps 2 and 3, rooms can still join meetings hosted on your own Zoom account; they just cannot join meetings hosted by other companies.

4 Put the credentials on each device

- **Make the file.** Copy `deploy/rooms/zoom-credentials.example.json` to `zoom-credentials.json` and fill in the six values: the step 1 client id and secret, the step 2 account id, client id and secret, and that room's user as `room_user`. Each room gets its own file because `room_user` differs.
- **Copy it to the device**, for example `scp zoom-credentials.json pi@crystal-meet-room-1.local:~/`.
- **A device you are installing now.** Add `--zoom-credentials ~/zoom-credentials.json` to the installer command from the setup guide.
- **A device that is already installed.** Type `sudo install -o pi -g pi -m 600 ~/zoom-credentials.json /etc/croom/zoom-credentials.json` (your username instead of pi), then `sudo systemctl restart croom`.
- **Check it.** Type `/opt/croom/venv/bin/croom --check-zoom -c /etc/croom/config.yaml`. It reads:

```
Zoom Meeting SDK: app AbC123xYz, signature minted (Zoom checks it only when a meeting is joined)
Zoom account: server-to-server token obtained for account 9XyZ12ab
Zoom room user: room1@crystalpm.com, ZAK obtained (valid 2 hours)
Ready: this room should join meetings hosted by any Zoom account; the first outside-hosted meeting
proves it.
```

- **Delete the copy in your home folder.** Type `rm ~/zoom-credentials.json`.

You should now see the check end with "Ready".

5 Join a test meeting

- **Inside your account.** Have a colleague start a Zoom meeting and paste its invitation link, passcode included, into the room page's **Join with a link**, or book it on the room's calendar and press **Join now**. Zoom's meeting appears on the TV within about fifteen seconds.
- **Outside your account.** Join a meeting hosted by someone outside the company, or by a personal Zoom account. The room appears in the participant list under its own name, for example Room 1. If the host uses a waiting room, the room page says it is waiting until the host admits it. This step is the proof for the outside-host route, which has not yet been verified against a live meeting: if Zoom refuses with an authorization message even though the check passed, tell whoever set this up; the fallback is fetching the ZAK through the SDK app's own OAuth.
- **Controls.** From the table page press **Mute**, **Turn camera off**, then **Leave** and **Tap again to leave**.

You should now see both meetings join, the room named after itself in the outside one, and the TV return to the room's idle screen after leaving.

Troubleshooting

The check says "not configured"

It names the value that is missing or still a placeholder in `/etc/croom/zoom-credentials.json`. The four server-to-server values and `room_user` go together: fill in all of them or none.

The check says Zoom refused the credentials

The account id, client id or secret of the step 2 app is wrong, or the app is not activated. Copy them again from App Credentials and check the Activation page.

The check says no such user, or a scope is missing

The `room_user` must be an active user on this Zoom account. "Lacks the user token scope" means step 2's scope `user:read:token:admin` is missing: add it and activate the app again.

The room page says the meeting is not authorized

The meeting is hosted by another account and the room joined without its Zoom user. Fill in the server-to-server values and `room_user`, restart the service and run the check. If the check passes and Zoom still refuses, the outside-host route needs the fallback described in step 5.

Wrong passcode or meeting not found

Paste the full invitation link with its passcode part. A bare meeting id cannot open a passcode-protected meeting. Zoom's exact reason shows on the room page.

The TV stays dark and the page says the SDK did not load

The device could not reach `source.zoom.us`. Check the device's internet access and any web filter, then try again.

GOOD TO KNOW

The room joins meetings as its own Zoom user through Zoom's supported Meeting SDK, so it is never mistaken for a bot. To take that away, deactivate the two apps or the room users in Zoom.